

Weekly Report – AEM Workflow & Component Development

Overview

This week focused on advanced AEM development concepts including workflows, launchers, multifield components, dispatcher understanding, and AEM API objects. Since workflows and dispatcher were new concepts, initial time was spent understanding their architecture, execution flow, and practical usage before implementation work was started.

Implementation Details

1. AEM Workflows & Launchers

- Studied workflow architecture and execution flow in AEM.
- Learned about:
 - Workflow Models
 - Workflow Instances
 - Process & Participant Steps
 - Launchers
- Created and tested custom workflow models.
- Configured workflow launchers to trigger workflows automatically on content changes.

2. Multifield Components

- Worked on multifield dialogs in AEM components.
- Captured multiple entries dynamically from author dialog.
- Rendered multifield data using HTL loops.
- Tested JCR storage structure for multifield data.

3. ECMA Script

- Learned basics of ECMA scripting in workflows.
- Used scripts for accessing workflow payload and metadata.
- Tested workflow execution using ECMA script steps.

4. Dispatcher & AEM API Objects

- Studied Dispatcher request flow, caching, filters, and headers.
- Explored AEM API objects:
 - Resource
 - ResourceResolver
 - Node
 - Session
 - PageManager
 - QueryBuilder
- Practiced accessing repository content programmatically.

Testing & Results

- Workflow models executed successfully.
- Launchers triggered workflows correctly.
- Multifield components rendered dynamic content properly.
- ECMA scripts executed without errors.
- Dispatcher concepts and API object usage were validated through testing.

Challenges & Solutions

- Understanding workflow execution flow initially → resolved through default workflow analysis.
- Multifield data rendering issues → solved using HTL iteration.
- Dispatcher architecture understanding required additional documentation study.
- Differentiating AEM API objects → practiced with repository examples.

Next Steps

- Implement custom workflow process steps using Java.
- Optimize component structure and reusable dialogs.
- Work more with Sling Models and backend integrations.